

Stochastic Finance

Financial Mathematics for Data Science — Lecture Notes

Alvaro Mendez

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1 About these notes

1.1 Scope

These are lecture notes for *Stochastic Finance — Financial Mathematics for Data Science* (third-semester elective, Master in Physics of Data, University of Padova, instructor Prof. Claudio Fontana). The course is a first mathematical treatment of financial derivatives: how they are priced, how their risks are hedged, and what mathematical machinery (stochastic processes, no-arbitrage arguments, martingale representations) makes the treatment rigorous.

1.2 Reading the notes

The notes follow the classical continuous-time development:

- **Chapters 1–2:** economic and institutional background — what markets are, what derivatives are, what “hedging” and “arbitrage” mean.
- **Chapter 3:** the binomial (Cox–Ross–Rubinstein) model, the simplest self-consistent framework where every derivative can be priced and hedged by explicit construction.
- **Chapter 4:** the mathematical infrastructure — Brownian motion, Itô’s lemma, stochastic differential equations.
- **Chapter 5:** the Black–Scholes–Merton model and the celebrated pricing formula.
- **Chapter 6:** Monte-Carlo methods for path-dependent options and high-dimensional derivatives.
- **Chapter 7:** local volatility, stochastic volatility and calibration to option surfaces.
- **Chapter 8:** short-rate models and the modern multi-curve framework for interest-rate derivatives.

Every chapter uses the styled callout environments to separate definitions, theorems, worked examples and remarks from the surrounding prose.

1.3 Notation

- S_t : spot price of an underlying asset at time t .
- B_t : value of the risk-free money-market account, $dB_t = rB_t dt$.
- W_t : standard Brownian motion; $W_t^{\mathbb{Q}}$ under an equivalent measure $\mathbb{Q}$.
- $\mathbb{P}, \mathbb{Q}$: physical (statistical) and risk-neutral measures.
- $\mathcal{F}_t$: filtration modelling information available at time t .
- $H_t = (\alpha_t, \beta_t)$: self-financing trading strategy in units of asset and cash.
- V_t^H : value of the portfolio H at time t .
- $C(t, S; K, T)$: value of a European call with strike K and maturity T .
- $\sigma(S, t)$: local volatility; $\sigma^{BS}(K, T)$: Black–Scholes implied volatility.
- $\Delta, \Gamma, \Theta, \mathcal{V}, \rho$: the option Greeks.

1.4 Sources

These notes are a re-organised version of *Stochastic Finance* lecture slides by Prof. Claudio Fontana (University of Padova, 2025) with additional material from the standard textbooks:

- Tomas Björk, *Arbitrage Theory in Continuous Time*, 4th ed., Oxford, 2020.
- Marek Musiela & Marek Rutkowski, *Martingale Methods in Financial Modelling*, 2nd ed., Springer, 2005.
- John C. Hull, *Options, Futures, and Other Derivatives*, 11th ed., Pearson, 2021.

Every chapter closes with its own *Sources* section naming the specific slide decks and book chapters from which its content was drawn.

2 Financial markets

Before we start pricing derivatives it is worth understanding the setting in which they trade. This chapter is short on mathematics and long on institutional vocabulary: the players, the instruments, the sources of risk, and the language (“bid–ask”, “arbitrage”, “hedging”) that the rest of the course assumes.

2.1 What is a financial market?

Definition 2.1 (Financial market). A **financial market** is an organised venue where economic agents exchange contingent claims — contracts whose payoff depends on future values of certain observable random variables (prices, rates, credit events). The market’s function is to allocate risk and capital across time and across agents.

The venue can be centralised (a *stock exchange*), electronically distributed (an *ECN*), or bilateral over-the-counter (OTC). What all markets share is a mechanism for matching buyers to sellers and a settlement infrastructure that guarantees delivery.

2.2 Main sources of risk

Proposition 2.1 (Classical taxonomy of financial risk). *Financial risk factors are traditionally decomposed into:*

- **Market risk** — losses from changes in market variables (*equity prices, FX rates, interest rates, commodity prices*).
- **Credit risk** — losses from counterparty default or from a credit rating downgrade.
- **Liquidity risk** — losses from being unable to close a position at fair value.
- **Operational risk** — losses from failures in processes, people, or systems.
- **Model risk** — losses from a mathematical model being wrong.

Remark. The 2007–2008 crisis was in essence a compound event: model risk (mispriced CDOs) triggered credit risk (mortgage defaults) which cascaded via liquidity risk (frozen interbank markets). Modern risk management treats these categories as strongly coupled, not orthogonal.

2.3 Buyers, sellers, and market makers

Definition 2.2 (Market participants). Every trade has three natural roles:

- **Buyers and sellers** — agents with directional views or hedging needs.
- **Market makers** — dealers who continuously post two-sided quotes (bid and ask) and profit from the *spread*.
- **Clearing houses** — central counterparties that step between buyer and seller after the trade to

mutualise counterparty risk.

Example 2.1 (*Bid-ask spread on FX*)

For a highly liquid pair (EUR/USD) the bid-ask spread quoted to a professional counterparty is ~ 0.1 pip on a \$1M ticket, i.e. transaction cost $\sim \$10$. For a less liquid pair (USD/TRY at 3 a.m.) the spread might be 10 pips, i.e. \$1000. The spread is the price of *immediate* execution: patient traders can post limit orders at mid and hope to get filled.

2.4 Asset classes

A financial market is typically split into *asset classes*, each with its own conventions and risk factors:

- **Equities** — stocks and stock indices.
- **Fixed income** — bonds and interest-rate derivatives.
- **Foreign exchange (FX)** — currency spot and forwards.
- **Commodities** — physical commodities and their futures.
- **Credit** — CDS and cash bonds referencing corporate or sovereign issuers.

Each has its own pricing conventions (yield vs price for bonds, forward points for FX, basis quotes for commodities), but the mathematical machinery — Itô calculus, risk-neutral valuation — is universal.

2.5 The time value of money

The single most important simple idea in finance:

Definition 2.3 (Discount factor). The **discount factor** $P(t, T)$ is the value at time t of receiving 1 unit of currency with certainty at future time T . In a deterministic-rate world with continuously compounded rate r ,

$$P(t, T) = e^{-r(T-t)}.$$

Proposition 2.2 (Present value of a cash flow stream). *The present value at time 0 of a deterministic cash flow $(c_1, c_2, \dots, c_n)$ paid at times $(t_1, \dots, t_n)$ is*

$$PV_0 = \sum_{i=1}^n c_i P(0, t_i).$$

2.6 Arbitrage

The concept that structures every derivative-pricing argument in the rest of the course:

Definition 2.4 (Arbitrage). An **arbitrage** is a self-financing trading strategy H such that the initial portfolio value is zero, the final value is non-negative with probability one, and strictly positive with positive probability:

$$V_0^H = 0, \quad V_T^H \geq 0 \text{ a.s.}, \quad P(V_T^H > 0) > 0.$$

Theorem 2.3 (No-arbitrage principle). *In a well-functioning market, arbitrages cannot exist — or if they do, they are exploited so quickly by traders that they disappear on timescales below the observation resolution. Pricing models are required to be **arbitrage-free**: their prices must satisfy consistency relations that rule out riskless profit.*

Example 2.2 (*Cash-and-carry arbitrage*)

Consider an equity S paying no dividends and a forward contract F maturing at T with strike K . If $K < S_0 e^{rT}$, buy the forward, short-sell the equity, invest the proceeds at rate r , and at T take delivery of the forward using the invested cash. Net cash flow: $S_0 e^{rT} - K > 0$ — arbitrage. Absence of arbitrage forces the forward price to equal $F = S_0 e^{rT}$.

2.7 Hedging

Definition 2.5 (Hedging). **Hedging** is the practice of offsetting the risk of one position with another instrument whose payoff moves in the opposite direction. A perfect hedge produces a portfolio whose value is deterministic (zero variance) over some time horizon.

Hedging turns out to be the *dual* of pricing: as we will see in Chapter 5, the Black–Scholes price of a European option is exactly the cost of the self-financing portfolio that perfectly replicates its payoff at maturity.

2.8 Regulated versus OTC markets

Most equity trading happens on regulated exchanges; most FX, credit and interest-rate derivative trading historically happened over-the-counter, bilaterally. After 2008 the trend has been strongly toward central clearing and standardisation:

Remark. A CDS on a major sovereign now clears through ICE Clear Credit; a standardised interest-rate swap in the major currencies clears through LCH. Central clearing greatly reduces counterparty risk but concentrates it in the clearing house — a new form of systemic risk that regulators watch closely.

2.9 Exercises

Exercise 2.1

A stock is trading at $S_0 = 100$. The 3-month risk-free rate is $r = 5\%$ (continuously compounded). What is the arbitrage-free forward price for delivery in 3 months?

Solution

$$F = S_0 e^{rT} = 100 \cdot e^{0.05 \cdot 0.25} \approx 101.26.$$

Exercise 2.2

The stock pays a discrete dividend D two months from now. How does the forward price change?

Solution

$F = (S_0 - PV(D))e^{rT} = (S_0 - De^{-rT_D})e^{rT}$ where $T_D = 2/12$. The forward is *lower* because the buyer of the forward does not receive the dividend, so its value must reflect the missed cash flow.

Exercise 2.3

Two zero-coupon bonds trade in the market: one maturing in 1 year at price $B_1 = 98$ and one in 2 years at $B_2 = 95$. What are the implied 1-year and 2-year spot rates?

Solution

$r_1 = -\ln(B_1/100) = -\ln 0.98 \approx 2.02\%$; $r_2 = -\frac{1}{2}\ln(B_2/100) = -\frac{1}{2}\ln 0.95 \approx 2.56\%$. The upward-sloping curve implies the market expects rates to rise (or a positive term premium).

2.10 Callouts

i Note

The no-arbitrage principle is an *idealisation*. Real markets show occasional small arbitrages (basis mispricings, ETF NAV drift) that persist for milliseconds; they are exploited by high-frequency market-makers, not by us.

💡 Tip

Whenever you're stuck on a pricing question, ask: "what portfolio has the same payoff?" If you can build a replicating portfolio, its cost *is* the price.

⚠ Warning

Discounting is at the *risk-free* rate only for deterministically-known cash flows. For risky flows the correct discount rate depends on the risk aversion of the investor and, in the risk-neutral world (Chapter 3), on the choice of *numéraire*.

2.11 Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 1 ([slides_1.pdf](#)). Hull, *Options, Futures, and Other Derivatives*, Chapter 1.

3 Financial derivatives

A derivative is a contract whose payoff at maturity depends on the future value of some other observable — the *underlying*. This chapter introduces the classical derivatives (forwards, futures, options, swaps) and the four fundamental parity relations that any consistent pricing framework must respect.

3.1 Classification

Definition 3.1 (Financial derivative). A **derivative** is a contract C with a specified payoff at (or before) a maturity date T , where the payoff is a function of the price path $\{S_t : 0 \leq t \leq T\}$ of an underlying S . The set of possible derivatives is enormous; the ones traded in liquid markets fall into a small number of families.

Remark (Terminology zoo). The two orthogonal axes of classification are the *exercise style* — European (only at T), American (any time $\leq T$), Bermudan (a finite set of dates) — and the *payoff structure* — vanilla (call or put), digital, barrier, Asian, look-back, etc. In these notes we focus on European vanilla until Chapter 6.

3.2 Forwards and futures

Definition 3.2 (Forward contract). A **forward** is a bilateral contract to buy (long) or sell (short) an underlying S at time T at a pre-agreed *forward price* F . The payoff at T for the long side is

$$V_T^{\text{fwd}} = S_T - F.$$

Proposition 3.1 (Arbitrage-free forward price). *Under no-arbitrage in a market with constant risk-free rate r and no dividends,*

$$F = S_0 e^{rT}.$$

With continuous dividends at rate q , $F = S_0 e^{(r-q)T}$; with FX, replace q by the foreign interest rate r_f .

Definition 3.3 (Futures contract). A **futures** is an exchange-traded standardised forward with *daily margining*: gains and losses on the position are settled in cash every day. Because of margining the futures price is not identical to the forward price when interest rates are stochastic, but the difference is typically small (< 1 basis point on liquid contracts).

3.3 Options

Definition 3.4 (European call and put). A **European call** on S with strike K and maturity T has payoff at T

$$C_T = (S_T - K)^+ = \max(S_T - K, 0).$$

A **European put** has payoff $P_T = (K - S_T)^+$.

A call gives the buyer the *right, not obligation* to buy S at price K on date T .

Example 3.1 (*Call option intuition*)

An investor buys a 6-month call on stock S with strike $K = 100$, paying premium $C_0 = 5$. At maturity, if $S_T = 120$ the investor exercises: pay $K = 100$, receive $S_T = 120$, net gain = $20 - 5 = 15$ (excluding time value of premium). If $S_T = 90$ the call expires worthless: loss = premium = 5.

Proposition 3.2 (Put–call parity). For European options on a non-dividend stock with strike K and maturity T ,

$$C_0 - P_0 = S_0 - Ke^{-rT}.$$

This is a model-free no-arbitrage identity: the two portfolios “long call + short put” and “long stock + short zero-coupon bond of face K ” have identical payoffs at T .

Remark. Put–call parity is one of the most useful identities in finance. It lets you infer put prices from call prices (or vice versa) without any pricing model, and it is the basis of *conversion* and *reversal* trading strategies used by market makers to lock in tiny mispricings.

3.4 Swaps

Definition 3.5 (Interest-rate swap). An **interest-rate swap** is a bilateral contract to exchange, on a schedule of dates $T_1 < T_2 < \dots < T_n$, a stream of fixed-rate cash flows for a stream of floating-rate cash flows on a notional amount N .

The fixed-leg cash flows are $NK\Delta_i$ at each T_i for some fixed rate K ; the floating-leg cash flows are $NL(T_{i-1}, T_i)\Delta_i$ where L is a market-observed reference rate (formerly LIBOR, now SOFR/€STR).

Proposition 3.3 (Swap fixed rate). The par swap rate — the fixed rate K^* that makes the initial value of the swap zero — is

$$K^* = \frac{P(0, T_0) - P(0, T_n)}{\sum_{i=1}^n \Delta_i P(0, T_i)},$$

where $P(0, T)$ is the discount factor to time T .

3.5 Payoff diagrams and structured products

Complex payoffs are conventionally described by their **payoff diagram** — payoff on the vertical axis vs S_T on the horizontal.

Example 3.2 (Bull spread)

A **bull spread** is long a call at strike K_1 and short a call at strike $K_2 > K_1$, both with maturity T . Its payoff is

$$V_T = (S_T - K_1)^+ - (S_T - K_2)^+ = \min(\max(S_T - K_1, 0), K_2 - K_1).$$

The buyer profits from a modest rise in S but caps the upside above K_2 . Cheaper than a single call.

Example 3.3 (Straddle)

A **straddle** is long a call and a put at the same strike $K = S_0$. Payoff: $|S_T - S_0|$. Profits from any large move in either direction, regardless of sign — a pure *volatility* bet.

3.6 Payoff decomposition and static hedging

Theorem 3.4 (Carr–Madan formula). *Any twice-differentiable payoff $g(S_T)$ can be replicated by a portfolio of a bond, a forward, and calls/puts of all strikes:*

$$g(S_T) = g(F) + g'(F)(S_T - F) + \int_0^F g''(K)(K - S_T)^+ dK + \int_F^\infty g''(K)(S_T - K)^+ dK,$$

where F is any reference level (e.g. the forward price).

The Carr–Madan formula is the *fundamental theorem of option replication*: it tells you the option surface is a *complete* basis for payoff functions. This is the mathematical reason why variance swaps and other volatility products can be replicated by static portfolios of vanilla calls and puts.

3.7 Option Greeks (preview)

The sensitivities of a derivative price to its inputs are called *Greeks*:

- $\Delta = \partial V / \partial S$: sensitivity to spot.
- $\Gamma = \partial^2 V / \partial S^2$: convexity in spot.
- $\Theta = -\partial V / \partial t$: time decay.
- $\mathcal{V} = \partial V / \partial \sigma$: vega, sensitivity to volatility.
- $\rho = \partial V / \partial r$: sensitivity to interest rate.

Greeks are what traders quote and hedge, not prices. Chapter 5 computes them in closed form for the Black–Scholes model.

3.8 Exercises

Exercise 3.1

Show that the payoff of a long forward at price F equals the payoff of a long call minus a long put at strike F : $(S_T - F) = (S_T - F)^+ - (F - S_T)^+$.

Solution

Case 1: $S_T > F$. Left: $S_T - F > 0$. Right: $(S_T - F) - 0 = S_T - F$. Case 2: $S_T < F$. Left: $S_T - F < 0$. Right: $0 - (F - S_T) = S_T - F$. So the identity holds for all S_T . Rearranging is put-call parity in payoff form.

Exercise 3.2

A European call with strike $K = 100$ and maturity $T = 1$ year trades at $C_0 = 8$; the corresponding put trades at $P_0 = 5$. The spot is $S_0 = 102$. What is the implied risk-free rate?

Solution

Put-call parity: $C_0 - P_0 = S_0 - Ke^{-rT}$, so $8 - 5 = 102 - 100e^{-r}$, giving $e^{-r} = 0.99$, i.e. $r \approx 1.005\%$.

Exercise 3.3

Use the Carr–Madan formula to replicate the log payoff $\ln(S_T/F)$ with a portfolio of calls and puts.

Solution

$g(S) = \ln(S/F)$: $g(F) = 0$, $g'(F) = 1/F$, $g''(K) = -1/K^2$. The replication is

$$\ln(S_T/F) = \frac{S_T - F}{F} - \int_0^F \frac{(K - S_T)^+}{K^2} dK - \int_F^\infty \frac{(S_T - K)^+}{K^2} dK.$$

This is the *variance-swap replication* formula: the log payoff is the crucial primitive in variance-swap pricing.

3.9 Callouts

i Note

Put-call parity is *model-free*: it holds whatever the true price process. Every option-pricing model must respect it, and this is a useful sanity check on numerical implementations.

💡 Tip

When designing a structured payoff, always draw the payoff diagram first. If the diagram looks “kinked” you need vanilla options; if it looks “smooth” you need a continuum of strikes.

⚠ Warning

American-style options cannot be replicated by static portfolios: their exercise boundary is stochastic. Chapters 3 and 5 briefly discuss the free-boundary problem for American options.

! Important

Options are quoted in *implied volatility*, not price. When a trader says “vol is up 2 vols” she means the Black–Scholes IV has moved by 200 bp — a bigger deal than a \$2 change in spot.

3.10 Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 2 ([slides_2.pdf](#)). Hull, *Options, Futures, and Other Derivatives*, Chapters 2, 9. Björk, *Arbitrage Theory in Continuous Time*, Chapter 6.

4 The binomial model

The simplest self-consistent framework in derivative pricing is the Cox–Ross–Rubinstein (CRR) binomial tree. It compresses the entire logic of no-arbitrage pricing into a discrete-time discrete-space model where every step can be checked by hand — and yet in the appropriate limit it recovers Black–Scholes. This chapter derives the one-step model, extends it to n steps, and shows how it converges to continuous time.

4.1 One-step binomial model

Definition 4.1 (One-step binomial market). A **one-step binomial market** has:

- One risky asset with initial price S_0 and time- T price $S_T \in \{S_0u, S_0d\}$, where $u > 1 > d > 0$.
- One risk-free asset (the *money-market account*) with price $B_t = e^{rt}$ (so $B_0 = 1$ and $B_T = e^{rT}$).
- A physical probability $p_u = P(S_T = S_0u)$ of the “up” state; $p_d = 1 - p_u$.

Proposition 4.1 (No-arbitrage in one step). *The market is arbitrage-free iff $d < e^{rT} < u$.*

Proof. If $e^{rT} \leq d$, borrow S_0 and buy the stock: at T pay back $S_0e^{rT} \leq S_0d \leq S_T$, keep a non-negative surplus with strictly positive probability — arbitrage. Symmetrically for $e^{rT} \geq u$. Both inequalities together rule out sign-definite outcomes.

4.2 Replicating a payoff

Suppose we want to price a derivative with payoff $V_T = f(S_T) \in \{f(S_0u), f(S_0d)\} = \{V_u, V_d\}$.

Theorem 4.2 (Replication in the binomial model). *There exist unique $\Delta \in \mathbb{R}$ and $\beta \in \mathbb{R}$ such that the portfolio “ Δ shares of S + β units of bond” matches the payoff in both states:*

$$\Delta = \frac{V_u - V_d}{S_0(u - d)}, \quad \beta = \frac{uV_d - dV_u}{u - d} e^{-rT}.$$

Hence the no-arbitrage price of the derivative at time 0 is

$$V_0 = \Delta S_0 + \beta = e^{-rT} [qV_u + (1 - q)V_d], \quad q = \frac{e^{rT} - d}{u - d} \in (0, 1).$$

Remark (Physical meaning). The quantity q has no relation to the physical probability p_u ; it is a synthetic weight determined entirely by r, u, d . It is called the **risk-neutral probability** of the up state, and $V_0 = e^{-rT} \mathbb{E}^Q[V_T]$ where $\mathbb{Q}$ is the measure that assigns probability q to the up state.

4.3 Risk-neutral valuation

The one-step result generalises to any dimension and any tree structure:

Theorem 4.3 (Fundamental theorem of asset pricing (finite version)). *A finite-tree market is arbitrage-free iff there exists a probability measure $\mathbb{Q}$ (equivalent to $\mathbb{P}$) under which the discounted asset price $\tilde{S}_t = e^{-rt} S_t$ is a martingale:*

$$\mathbb{E}^{\mathbb{Q}}[\tilde{S}_{t+1} | \mathcal{F}_t] = \tilde{S}_t.$$

*The market is **complete** — every payoff can be replicated — iff $\mathbb{Q}$ is unique.*

Corollary 4.4 (Risk-neutral pricing formula). *In an arbitrage-free complete market, the no-arbitrage price at time t of a derivative with payoff V_T is*

$$V_t = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[V_T | \mathcal{F}_t].$$

4.4 Multi-step binomial model

Iterate the one-step model n times. The tree has 2^n paths but only $n+1$ terminal nodes (the number of up moves determines S_T). At each internal node the local replication argument applies with the *same* (u, d, r) , so backwards induction gives the price at every node.

Proposition 4.5 (n -step price of a European derivative). *For a European payoff $V_T = f(S_T)$ in an n -step CRR tree with step size Δt ,*

$$V_0 = e^{-rT} \sum_{k=0}^n \binom{n}{k} q^k (1-q)^{n-k} f(S_0 u^k d^{n-k}),$$

where $q = (e^{r\Delta t} - d)/(u - d)$ is the one-step risk-neutral probability.

4.5 American options and the free-boundary problem

For an **American** option the holder can exercise early. Pricing requires backwards induction with a comparison at each node between the immediate exercise value and the continuation value:

$$V(S, t) = \max(f(S), e^{-r\Delta t}[qV^u + (1-q)V^d]).$$

Example 4.1 (Early exercise of an American put)

For an American put on a non-dividend stock with strike K : as spot falls below a stochastic *early-exercise boundary* $B(t)$, immediate exercise beats holding. The boundary shrinks toward K as $t \rightarrow T$ (holding until maturity becomes safer as time runs out) and away from K at $t = 0$. The exact form requires solving a free-boundary PDE; the binomial tree is the standard numerical method.

Remark. An American *call* on a non-dividend stock is never exercised early — you always prefer the time value of the strike (invest K at rate r) plus the option to walk away. This is Merton's classical no-early-exercise theorem.

4.6 Convergence to Black–Scholes

Take the step size $\Delta t = T/n \rightarrow 0$ and calibrate

$$u = e^{\sigma\sqrt{\Delta t}}, \quad d = e^{-\sigma\sqrt{\Delta t}} = 1/u.$$

Then the log return per step has variance $\sigma^2\Delta t$, so the cumulative log return $\ln(S_T/S_0)$ has variance σ^2T in the limit — matching the Black–Scholes model.

Theorem 4.6 (CRR converges to Black–Scholes). *Let $C_0^{(n)}$ denote the CRR call price with n steps as above. Then*

$$\lim_{n \rightarrow \infty} C_0^{(n)} = S_0 \Phi(d_1) - Ke^{-rT} \Phi(d_2),$$

the Black–Scholes formula (Chapter 5), where Φ is the standard-normal CDF.

Proof idea. Invoke the central limit theorem for the n -step binomial: the log price converges in distribution to a Brownian motion with drift $r - \sigma^2/2$ and diffusion σ . Discounted-expectation of $(S_T - K)^+$ under the resulting log-normal distribution gives the BS formula.

4.7 Exercises

Exercise 4.1

A one-step binomial model has $S_0 = 100$, $u = 1.2$, $d = 0.9$, $r = 5\%$, $T = 1$ year. Price a European call with strike $K = 100$.

Solution

Risk-neutral probability: $q = (e^{0.05} - 0.9)/(1.2 - 0.9) \approx 0.5085$. Payoffs: $V_u = 20$, $V_d = 0$. Price: $V_0 = e^{-0.05}(0.5085 \cdot 20 + 0.4915 \cdot 0) \approx 9.67$. Delta: $\Delta = 20/(100 \cdot 0.3) \approx 0.667$.

Exercise 4.2

Show that for the CRR calibration $u = 1/d$, the risk-neutral probability converges to $1/2$ as $\Delta t \rightarrow 0$ if the drift under $\mathbb{Q}$ is zero. Comment on the implication.

Solution

$q = (e^{r\Delta t} - d)/(u - d)$. Taylor-expanding: $e^{r\Delta t} \approx 1 + r\Delta t$, $u \approx 1 + \sigma\sqrt{\Delta t} + \sigma^2\Delta t/2$, $d \approx 1 - \sigma\sqrt{\Delta t} + \sigma^2\Delta t/2$. Leading order: $q \approx 1/2 + (r - \sigma^2/2)\sqrt{\Delta t}/(2\sigma)$. As $\Delta t \rightarrow 0$ we recover $q = 1/2$; the drift correction lives in higher-order terms.

4.8 Callouts

i Note

The binomial model is *finite* — no measure theory needed to state the fundamental theorem of asset pricing. This is why it is taught first: all the abstractions of Chapter 4 have their finite analogues here.

 Tip

For pricing an American option numerically, the binomial tree is very hard to beat: it is trivial to code, easy to debug, and (with an appropriate number of steps) accurate to two decimal places. Only for high-dimensional or path-dependent options do more advanced techniques win.

 Warning

The convergence of CRR to BS is $O(1/n)$ but *oscillatory*: prices with different parities of n approach BS from opposite sides. Averaging two adjacent n values (Richardson extrapolation) gives $O(1/n^2)$ convergence.

 Important

Risk-neutral probability is a mathematical trick, not a prediction of what will happen. Do not tell a client that “the market thinks the up move has probability q ”.

4.9 Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 3 ([slides_3.pdf](#)). Björk, *Arbitrage Theory in Continuous Time*, Chapter 2. Hull, *Options, Futures, and Other Derivatives*, Chapters 13–14. Original references: Cox, Ross & Rubinstein, *Option pricing: a simplified approach*, J. Fin. Econ. **7**, 229 (1979).

5 Brownian motion and stochastic calculus

Continuous-time derivative pricing requires a continuous-time model of the underlying. The paradigm is *Brownian motion*, and the calculus that handles integrals against it is Itô's calculus. This chapter is the mathematical backbone of everything from here on.

5.1 Brownian motion

Definition 5.1 (Standard Brownian motion). A **standard Brownian motion** $\{W_t\}_{t \geq 0}$ on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$ is a stochastic process such that:

1. $W_0 = 0$ a.s.
2. Increments $W_t - W_s$ are independent of $\mathcal{F}_s$ for $s < t$.
3. $W_t - W_s \sim \mathcal{N}(0, t - s)$ for $s \leq t$.
4. $t \mapsto W_t$ is continuous a.s.

Proposition 5.1 (Basic properties of Brownian motion). *For standard Brownian motion:*

- **Martingale.** $\mathbb{E}[W_t | \mathcal{F}_s] = W_s$ for $s \leq t$.
- **Quadratic variation.** For any $t > 0$, $[W]_t = t$: the (bracket) accumulates variance at rate one.
- **Nowhere differentiable.** With probability one, $t \mapsto W_t$ is continuous but not differentiable at any point.
- **Scaling invariance.** For any $c > 0$, $\{c^{-1/2}W_{ct}\}_{t \geq 0}$ is again a standard BM.
- **Reflection principle.** For any $a > 0$, $P(\max_{s \leq t} W_s \geq a) = 2P(W_t \geq a)$.

Remark (Physical origin). Brownian motion was named for the pollen-grain jiggling observed by botanist Robert Brown in 1827. Einstein (1905) derived the diffusion equation from the same phenomenon and set the microscopic mean-square displacement at $\langle x^2 \rangle \sim t$. Kolmogorov (1933) axiomatised the mathematical object; Norbert Wiener (1923) constructed it — hence the name “Wiener process”.

5.2 Filtrations and martingales

Definition 5.2 (Filtration). A **filtration** $\{\mathcal{F}_t\}_{t \geq 0}$ is an increasing family of σ -algebras: $\mathcal{F}_s \subseteq \mathcal{F}_t$ for $s \leq t$. Intuitively $\mathcal{F}_t$ contains all information observable up to time t . A process X is **adapted** to $\{\mathcal{F}_t\}$ if X_t is $\mathcal{F}_t$ -measurable for every t .

Definition 5.3 (Martingale). An integrable adapted process X is a **martingale** with respect to $\{\mathcal{F}_t\}$ if $\mathbb{E}[X_t | \mathcal{F}_s] = X_s$ for all $s \leq t$. A **supermartingale** has $\leq$, a **submartingale** has $\geq$.

Martingales formalise the “fair game” property: the best forecast of a martingale, given today's information, is its current value.

Example 5.1 (Standard martingales from BM)

The processes W_t , $W_t^2 - t$, and $\exp(\theta W_t - \theta^2 t/2)$ (the *exponential martingale*) are all martingales in the filtration generated by W . These three are the building blocks of most calculations.

5.3 Itô integral

The Itô integral $\int_0^t H_s dW_s$ generalises the Riemann integral to integrators of unbounded variation, exploiting the fact that Brownian motion has finite quadratic variation.

Definition 5.4 (Itô integral for simple integrands). For a piecewise-constant integrand $H_s = \sum_k h_k \mathbf{1}_{(t_k, t_{k+1}]}(s)$ with h_k $\mathcal{F}_{t_k}$ -measurable, the Itô integral is

$$\int_0^t H_s dW_s = \sum_k h_k (W_{t_{k+1} \wedge t} - W_{t_k \wedge t}).$$

The definition extends by isometry to general square-integrable adapted H .

Proposition 5.2 (Itô isometry). For H square-integrable and adapted,

$$\mathbb{E} \left[\left(\int_0^t H_s dW_s \right)^2 \right] = \mathbb{E} \left[\int_0^t H_s^2 ds \right].$$

The Itô integral is a martingale with zero mean.

5.4 Itô's lemma

The chain rule for stochastic calculus:

Theorem 5.3 (Itô's lemma). Let X_t satisfy the SDE $dX_t = \mu_t dt + \sigma_t dW_t$, and $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ be $C^{1,2}$. Then

$$df(t, X_t) = (\partial_t f + \mu_t \partial_x f + \frac{1}{2} \sigma_t^2 \partial_x^2 f) dt + \sigma_t \partial_x f dW_t.$$

The correction term $\frac{1}{2} \sigma_t^2 \partial_x^2 f$ is the signature of stochastic calculus: it arises because $(dW_t)^2 = dt$ at second order.

Example 5.2 (Geometric Brownian motion)

Applying Itô to $S_t = S_0 \exp(X_t)$ with $dX_t = \mu dt + \sigma dW_t$: with $f(x) = e^x$ we get

$$dS_t = S_t(\mu + \frac{1}{2}\sigma^2)dt + S_t \sigma dW_t.$$

The natural log SDE has drift μ ; the level SDE has drift $\mu + \sigma^2/2$ — the Itô correction.

5.5 SDEs and their solutions

Definition 5.5 (Stochastic differential equation). An **Itô SDE** with drift $b : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ and diffusion

$\sigma : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ is

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 = x_0.$$

A **strong solution** is an $\mathcal{F}_t$ -adapted process X_t satisfying the integral form of the SDE.

Theorem 5.4 (Existence and uniqueness). *If b and σ are globally Lipschitz in x (uniformly in t) with linear growth, then the SDE has a unique strong solution on $[0, T]$.*

Example 5.3 (Ornstein–Uhlenbeck process)

$dX_t = \kappa(\bar{x} - X_t) dt + \sigma dW_t$: mean-reverting Gaussian process. Explicit solution:

$$X_t = X_0 e^{-\kappa t} + \bar{x}(1 - e^{-\kappa t}) + \sigma \int_0^t e^{-\kappa(t-s)} dW_s.$$

Asymptotic distribution $\mathcal{N}(\bar{x}, \sigma^2/(2\kappa))$. The OU process is the workhorse of short-rate modelling (Chapter 8) and stochastic volatility (Chapter 7).

5.6 Girsanov's theorem

The bridge from the physical measure $\mathbb{P}$ to a risk-neutral measure $\mathbb{Q}$ is a *measure change* engineered by Girsanov:

Theorem 5.5 (Girsanov). *Let θ_t be adapted with $\mathbb{E}[\exp(\frac{1}{2} \int_0^T \theta_s^2 ds)] < \infty$ (Novikov's condition). Define the exponential martingale*

$$Z_t = \exp \left(- \int_0^t \theta_s dW_s - \frac{1}{2} \int_0^t \theta_s^2 ds \right), \quad \mathbb{E}[Z_T] = 1.$$

Then $d\mathbb{Q}/d\mathbb{P} = Z_T$ defines a probability measure $\mathbb{Q}$ equivalent to $\mathbb{P}$, and $\tilde{W}_t = W_t + \int_0^t \theta_s ds$ is a standard $\mathbb{Q}$ -Brownian motion.

Corollary 5.6 (Change of drift for GBM). *If S follows $dS_t = \mu S_t dt + \sigma S_t dW_t$ under $\mathbb{P}$, then under $\mathbb{Q}$ defined by $\theta = (\mu - r)/\sigma$ the drift becomes r :*

$$dS_t = r S_t dt + \sigma S_t d\tilde{W}_t,$$

i.e. S is drifted at the risk-free rate — the essence of risk-neutral valuation.

5.7 Fundamental theorem of asset pricing

Theorem 5.7 (FTAP (continuous-time)). *An arbitrage-free market with numéraire B admits an equivalent measure $\mathbb{Q}$ under which every discounted asset S/B is a $\mathbb{Q}$ -martingale. Completeness of the market corresponds to uniqueness of $\mathbb{Q}$.*

Remark. This is the deep, continuous-time analogue of the finite-tree FTAP of Chapter 3. Its proof uses Girsanov to convert away the drift and semi-martingale representation theorems to characterise the set

of equivalent measures.

5.8 Exercises

Exercise 5.1

Show that $M_t = W_t^2 - t$ is a martingale by direct computation.

Solution

For $s < t$: $\mathbb{E}[W_t^2 | \mathcal{F}_s] = \mathbb{E}[(W_s + (W_t - W_s))^2 | \mathcal{F}_s] = W_s^2 + 2W_s\mathbb{E}[W_t - W_s | \mathcal{F}_s] + \mathbb{E}[(W_t - W_s)^2] = W_s^2 + 0 + (t - s)$. So $\mathbb{E}[M_t | \mathcal{F}_s] = W_s^2 + (t - s) - t = W_s^2 - s = M_s$.

Exercise 5.2

Apply Itô's lemma to compute $d(tW_t)$. Deduce that $\int_0^t s dW_s = tW_t - \int_0^t W_s ds$.

Solution

$f(t, x) = tx$: $\partial_t f = x$, $\partial_x f = t$, $\partial_x^2 f = 0$. So $d(tW_t) = W_t dt + t dW_t$. Rearranging: $\int_0^t s dW_s = tW_t - \int_0^t W_s ds$. This is integration by parts under the Itô rule.

Exercise 5.3

Compute the mean and variance of the Ornstein–Uhlenbeck solution starting from $X_0 = x_0$. Verify convergence to the stationary distribution.

Solution

$\mathbb{E}[X_t] = x_0 e^{-\kappa t} + \bar{x}(1 - e^{-\kappa t})$; $\text{Var}(X_t) = \sigma^2(1 - e^{-2\kappa t})/(2\kappa)$. As $t \rightarrow \infty$: $\mathbb{E} \rightarrow \bar{x}$, $\text{Var} \rightarrow \sigma^2/(2\kappa)$. Gaussian stationary distribution $\mathcal{N}(\bar{x}, \sigma^2/(2\kappa))$.

5.9 Callouts

Note

Itô's lemma is *not* a chain rule of ordinary calculus with extra terms. The extra $\frac{1}{2}\sigma^2\partial_x^2 f dt$ arises because dW has magnitude $\sqrt{dt}$, so second-order terms contribute at first order.

Tip

When solving an SDE by hand, first try to guess a function $Y_t = f(t, X_t)$ that satisfies a simpler SDE (constant coefficients, or deterministic). This turns the problem into a straightforward integration.

Warning

Stratonovich integrals $(X \circ dW)$ and Itô integrals $(X dW)$ give different values for stochastic integrals. Finance uses Itô convention; classical physics often uses Stratonovich. Be careful when reading across disciplines.

 Caution

The convergence of a Monte-Carlo Euler scheme for an SDE is *weak*: sample paths do not converge to the true path, only their laws do. If you need pathwise convergence, use a Milstein or higher-order scheme.

5.10 Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 4 ([slides_4.pdf](#)). Björk, *Arbitrage Theory in Continuous Time*, Chapters 4–5. Karatzas & Shreve, *Brownian Motion and Stochastic Calculus*, 2nd ed., Springer (1998).

6 The Black–Scholes–Merton model

With Itô calculus in hand we can now derive the Black–Scholes–Merton (BSM) pricing formula — the most influential single result in modern finance. This chapter derives it two ways (PDE via delta-hedging; risk-neutral via Girsanov), computes the Greeks, and discusses the model’s spectacular successes and equally spectacular failures.

6.1 The BSM assumptions

Definition 6.1 (BSM market). The **Black–Scholes–Merton market** consists of:

- One risky asset with dynamics $dS_t = \mu S_t dt + \sigma S_t dW_t$ under $\mathbb{P}$ (**geometric Brownian motion**), constant drift μ and volatility $\sigma > 0$.
- One risk-free money-market account $B_t = e^{rt}$ with constant r .
- No dividends, no transaction costs, no taxes, unlimited borrowing and lending at rate r , and continuous trading.

6.2 Delta-hedging derivation

Let $V(t, S)$ be the value of a self-financing portfolio that replicates a derivative with terminal payoff $V(T, S_T) = f(S_T)$.

Theorem 6.1 (Black–Scholes PDE). *The value $V(t, S)$ of a European derivative in the BSM market satisfies the parabolic PDE*

$$\partial_t V + rS \partial_S V + \frac{1}{2} \sigma^2 S^2 \partial_S^2 V - rV = 0, \quad V(T, S) = f(S).$$

Proof. Consider the portfolio $\Pi = V - \Delta S$ with $\Delta = \partial_S V$. Applying Itô’s lemma to $V(t, S_t)$ and using $dS = \mu S dt + \sigma S dW$,

$$dV = (\partial_t V + \mu S \partial_S V + \frac{1}{2} \sigma^2 S^2 \partial_S^2 V) dt + \sigma S \partial_S V dW.$$

Choosing $\Delta = \partial_S V$ kills the dW term in $d\Pi$. The remaining portfolio must earn the risk-free rate (no-arbitrage), $d\Pi = r\Pi dt$. Rearranging gives the PDE.

6.3 Black–Scholes formula (call)

Theorem 6.2 (Black–Scholes formula). *The value at time t of a European call with strike K and maturity T in the BSM market is*

$$C(t, S) = S \Phi(d_1) - K e^{-r(T-t)} \Phi(d_2),$$

where Φ is the standard normal CDF and

$$d_{1,2} = \frac{\ln(S/K) + (r \pm \frac{1}{2}\sigma^2)(T-t)}{\sigma\sqrt{T-t}}.$$

Proof. Under the risk-neutral measure $\mathbb{Q}$, $dS = rS dt + \sigma S d\tilde{W}$, so $S_T = S_t \exp((r - \sigma^2/2)(T-t) + \sigma(\tilde{W}_T - \tilde{W}_t))$. The pricing formula $C = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[(S_T - K)^+ | \mathcal{F}_t]$ evaluates by a Gaussian integral to the stated form.

Corollary 6.3 (Put by parity). *By put–call parity (Chapter 2):*

$$P(t, S) = K e^{-r(T-t)} \Phi(-d_2) - S \Phi(-d_1).$$

6.4 The Greeks

Proposition 6.4 (Greeks for a European call). *For the BSM call price above, the sensitivities are:*

- $\Delta = \partial_S C = \Phi(d_1)$
- $\Gamma = \partial_S^2 C = \phi(d_1) / (S\sigma\sqrt{T-t})$
- $\Theta = \partial_t C = -\frac{S\phi(d_1)\sigma}{2\sqrt{T-t}} - rK e^{-r(T-t)} \Phi(d_2)$
- $\mathcal{V} = \partial_\sigma C = S\phi(d_1)\sqrt{T-t}$
- $\rho = \partial_r C = K(T-t)e^{-r(T-t)}\Phi(d_2)$

where $\phi = \Phi'$ is the standard normal PDF.

Remark (Greeks in practice).

- Delta is between 0 and 1 for a call; a trader who is long 100 shares hedges by selling $100/\Delta$ calls.
- Gamma peaks at the strike near maturity and is what makes delta-hedging expensive: rebalancing costs scale as $\int_0^T \frac{1}{2}\sigma^2 S_t^2 \Gamma dt$ in expectation.
- Vega measures exposure to volatility misspecification; sign is always positive for vanilla options.
- Theta is the “rent” the option holder pays for the time value of the position.

6.5 Implied volatility and the smile

The BSM formula is invertible in σ : given a market price C^{mkt} , there is a unique σ^{IV} such that $C^{BSM}(\sigma^{IV}) = C^{mkt}$.

Definition 6.2 (Implied volatility). The **implied volatility** $\sigma^{IV}(K, T)$ of an option with strike K and maturity T is the σ that makes the BSM formula match the market price.

If the BSM model were correct, σ^{IV} would be constant across strikes and maturities. In practice it is *not*:

Example 6.1 (The volatility smile)

For equity index options (SPX, EURO STOXX 50), the implied volatility surface shows a *skew*: OTM puts have higher IV than ATM options, which have higher IV than OTM calls. The shape is roughly a smile for FX and a skew for equities. This is direct evidence that the BSM assumptions — constant σ , log-normal S — are wrong.

The volatility surface is the central object of modern derivative markets. Chapter 7 studies models that reproduce it.

6.6 Model successes and failures

Remark (What BSM got right). BSM's contribution is not the formula per se but the *methodology*: relative pricing by no-arbitrage replication. Every derivative-pricing model in the last 50 years is built on this foundation. The Black–Scholes formula itself remains the universal *quotation* mechanism (via implied volatility), even though no professional trader believes the model.

Remark (What BSM got wrong).

- **Fat tails.** Equity log-returns have kurtosis > 3 : extreme moves happen more often than log-normality predicts.
- **Stochastic volatility.** σ itself is a random variable that clusters in time (GARCH effects, ARCH regressions).
- **Jumps.** Real prices jump — corporate news, macro announcements — and continuous diffusion cannot capture that.
- **Correlation risk.** BSM prices individual options; the correlation between options of different strikes matters for structured products and is not in the model.

The 1987 crash, the 1998 LTCM collapse, and the 2008 crisis all involved losses that the standard BSM framework significantly underestimated.

6.7 Feynman–Kac and pricing PDEs

The link between PDE and probabilistic pricing is the Feynman–Kac theorem:

Theorem 6.5 (Feynman–Kac (financial version)). *Let $u(t, x)$ solve the parabolic PDE*

$$\partial_t u + \frac{1}{2} \sigma^2 x^2 \partial_x^2 u + r x \partial_x u - r u = 0, \quad u(T, x) = f(x).$$

Then

$$u(t, x) = e^{-r(T-t)} \mathbb{E}^{\mathbb{Q}}[f(X_T) \mid X_t = x],$$

where $dX_s = rX_s ds + \sigma X_s d\tilde{W}_s$ under $\mathbb{Q}$.

Feynman–Kac is the two-way street between the PDE and the risk-neutral expectation: you can solve either representation, whichever is more convenient.

6.8 Exercises

Exercise 6.1

Show that the BSM call price satisfies $C \rightarrow \max(S - Ke^{-r(T-t)}, 0)$ as $\sigma \rightarrow 0$, and $C \rightarrow S$ as $\sigma \rightarrow \infty$. Interpret physically.

Solution

$\sigma \rightarrow 0$: $d_1, d_2 \rightarrow \pm\infty$ depending on the sign of $\ln(S/K) + r(T-t)$. If $S > Ke^{-r(T-t)}$, $\Phi(d_1) \rightarrow 1$, $\Phi(d_2) \rightarrow 1$: $C \rightarrow S - Ke^{-r(T-t)}$. If $S < Ke^{-r(T-t)}$, both $\rightarrow 0$: $C \rightarrow 0$. Union: intrinsic value.
 $\sigma \rightarrow \infty$: $d_1 \rightarrow \infty$ and $d_2 \rightarrow -\infty$: $\Phi(d_1) \rightarrow 1$, $\Phi(d_2) \rightarrow 0$: $C \rightarrow S$. Physical interpretation: infinite volatility makes every strike equally likely, so the call is worth its underlying (the strike is worth nothing in expectation).

Exercise 6.2

Derive the vega $\mathcal{V} = \partial_\sigma C$ by differentiating the BSM formula directly, and verify $\mathcal{V} = S\phi(d_1)\sqrt{T-t}$.

Solution

$\partial_\sigma d_1 = \sqrt{T-t} - d_1/\sigma$, $\partial_\sigma d_2 = -\sqrt{T-t} - d_2/\sigma$. Chain rule: $\partial_\sigma C = S\phi(d_1)\partial_\sigma d_1 - Ke^{-r(T-t)}\phi(d_2)\partial_\sigma d_2$. Using the identity $S\phi(d_1) = Ke^{-r(T-t)}\phi(d_2)$ (from the Gaussian densities and the definition of d_1, d_2), the two terms combine to $S\phi(d_1)\sqrt{T-t}$.

Exercise 6.3

A trader is long a call with $\Delta = 0.7$, $\Gamma = 0.02$. She wants to be delta-neutral. How many shares does she need to short? If the stock moves by $\Delta S = 1$, what is her P&L (ignoring theta)?

Solution

Short 0.7 shares per option. After a move of $\Delta S = 1$: the call gains approximately $\Delta \cdot \Delta S + \frac{1}{2}\Gamma(\Delta S)^2 = 0.7 + 0.01 = 0.71$. The short-stock position loses $0.7 \cdot 1 = 0.7$. Net P&L = $0.71 - 0.7 = 0.01$: positive because $\Gamma > 0$. Gamma is the source of the *positive convexity* of a long-option position.

6.9 Callouts

i Note

The Black–Scholes formula is one of the most-tested equations in finance. Any implementation that produces a positive price, positive vega, and reproduces put–call parity is likely correct; anything else is buggy.

💡 Tip

Implied volatility is *the* language of options trading. Prices vary with spot; IVs vary with fear. Learning to think in IV terms is the single biggest step from student to trader.

! Important

Delta-hedging is a *dynamic* strategy: the option seller must trade continuously to remain hedged. Discrete rebalancing introduces a P&L variance proportional to $\sigma^2 S^2 \Gamma^2 \Delta t$ — an unavoidable cost.

 Warning

Every BSM Greek assumes the model is right. In a stochastic-volatility world (Chapter 7), the “true” vega differs from the BSM vega, and the extra Greeks (vanna, volga, vomma) matter for FX and structured products.

6.10 Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 5 ([slides_5.pdf](#) and [bs.pdf](#)). Björk, *Arbitrage Theory in Continuous Time*, Chapters 7–8. Hull, *Options, Futures, and Other Derivatives*, Chapters 15–19. Original references: Black & Scholes, *The pricing of options and corporate liabilities*, J. Pol. Econ. **81**, 637 (1973); Merton, *Theory of rational option pricing*, Bell J. Econ. **4**, 141 (1973).

7 Monte Carlo methods for derivatives

The Black–Scholes PDE admits a closed-form solution only for the simplest payoffs on the simplest underlyings. For anything more realistic — path-dependent payoffs (Asian, look-back), high-dimensional baskets, or non-Gaussian dynamics — the only tractable route is Monte Carlo. This chapter is about how to do it well.

7.1 Why Monte Carlo?

The risk-neutral pricing formula $V_0 = e^{-rT} \mathbb{E}^{\mathbb{Q}}[f(\mathbf{S}_T)]$ is an expectation. Monte Carlo replaces it with a sample average:

Definition 7.1 (Monte Carlo price estimator). Given N i.i.d. samples $\mathbf{S}_T^{(1)}, \dots, \mathbf{S}_T^{(N)}$ drawn from the $\mathbb{Q}$ -distribution of the terminal state,

$$\widehat{V}_0 = e^{-rT} \frac{1}{N} \sum_{i=1}^N f(\mathbf{S}_T^{(i)}).$$

By the CLT, $\widehat{V}_0 \rightarrow V_0$ with error $O(\sigma_f/\sqrt{N})$, independent of the dimension of $\mathbf{S}$.

The rate $1/\sqrt{N}$ is the *dimension-independent* advantage of Monte Carlo over quadrature. For a 10-basket option, quadrature needs M^{10} points where M is per-dimension resolution; Monte Carlo needs $\sim 10^6$ for 2-3 significant digits regardless of the basket size.

7.2 Simulating GBM

For a single log-normal underlying under $\mathbb{Q}$, $S_T = S_0 \exp((r - \sigma^2/2)T + \sigma\sqrt{T}Z)$ with $Z \sim \mathcal{N}(0, 1)$. Simulation is trivial: draw Z once per path.

For a path-dependent option we need the whole trajectory. The Euler–Maruyama scheme with step Δt :

Definition 7.2 (Euler–Maruyama). For $dX_t = b(X_t)dt + \sigma(X_t)dW_t$ with $X_0 = x_0$, generate the discrete trajectory

$$X_{t+\Delta t} = X_t + b(X_t)\Delta t + \sigma(X_t)\sqrt{\Delta t}Z_t, \quad Z_t \sim \mathcal{N}(0, 1) \text{ i.i.d.}$$

For GBM, this is

$$S_{t+\Delta t} = S_t(1 + r\Delta t + \sigma\sqrt{\Delta t}Z_t).$$

Remark (Log-scheme). For GBM it is better to simulate $\ln S$ rather than S , so the scheme has *exact*

increments and works at any Δt :

$$\ln S_{t+\Delta t} = \ln S_t + (r - \sigma^2/2)\Delta t + \sigma\sqrt{\Delta t} Z_t.$$

This avoids the small- Δt bias of the level scheme.

7.3 Variance reduction

The Monte Carlo error is $\sigma_f/\sqrt{N}$. Halving the error means $4\times$ more samples — very expensive. **Variance reduction** techniques exploit structure of the problem to reduce σ_f instead.

7.3.1 Antithetic variates

Proposition 7.1 (Antithetic variates). *Simulate paths in pairs: $\mathbf{Z}^{(i)}$ and $-\mathbf{Z}^{(i)}$. The estimator*

$$\widehat{V}^{av} = \frac{1}{N} \sum_i \frac{1}{2} [f(\mathbf{S}_T(\mathbf{Z}^{(i)})) + f(\mathbf{S}_T(-\mathbf{Z}^{(i)}))]$$

is unbiased. If f is monotone in each coordinate of $\mathbf{Z}$, the pair has negative correlation and the variance is reduced by up to 50%.

7.3.2 Control variates

Proposition 7.2 (Control variates). *Suppose $Y = f(\mathbf{S}_T)$ is the target and $Y' = g(\mathbf{S}_T)$ is a variable with known expectation $\mathbb{E}^{\mathbb{Q}}[Y'] = \mu'$. Then for any β ,*

$$\widehat{V}^{cv} = \frac{1}{N} \sum_i [Y^{(i)} - \beta(Y'^{(i)} - \mu')]$$

is unbiased with variance $\text{Var}(\widehat{V}^{cv}) = \text{Var}(Y)(1 - \rho^2)$, minimised at $\beta^ = \text{Cov}(Y, Y') / \text{Var}(Y')$. If Y' is chosen close to Y ($\rho \approx 1$) the variance reduction is dramatic.*

Example 7.1 (Control variate for an Asian call)

For an arithmetic-average Asian call $\mathbb{E}^{\mathbb{Q}}[(\bar{S}_T - K)^+]$, the analogous geometric-average call has a closed form (log-normal average). Use it as Y' ; typical variance reduction factor $\sim 50\times$.

7.3.3 Importance sampling

Proposition 7.3 (Importance sampling). *Suppose the payoff f is non-zero only on a rare event ($f(\mathbf{S}_T) > 0$ has probability $\epsilon \ll 1$). Under a change of measure $\mathbb{Q} \rightarrow \tilde{\mathbb{Q}}$ with density $L = d\tilde{\mathbb{Q}}/d\mathbb{Q}$,*

$$\mathbb{E}^{\mathbb{Q}}[f(\mathbf{S}_T)] = \mathbb{E}^{\tilde{\mathbb{Q}}}[f(\mathbf{S}_T)/L(\mathbf{S}_T)].$$

If $\tilde{\mathbb{Q}}$ concentrates on the region where $f > 0$, the variance of the transformed estimator drops sharply.

Example 7.2 (Deep OTM options)

For a deep OTM call, $q = P^{\mathbb{Q}}(S_T > K)$ is tiny; the Monte Carlo estimator has variance $\sim K^2 q$. Shift the drift of the log-price by $\mu \rightarrow \mu + \delta$ under an importance measure — this brings more sample paths into the money and can reduce variance by orders of magnitude.

7.4 Low-discrepancy sequences

Sobol' or Halton sequences are deterministic sequences whose empirical distribution is *more uniform* than random samples. Using them in place of pseudo-random numbers gives *quasi-Monte Carlo* (QMC):

Theorem 7.4 (QMC error bound (Koksma–Hlawka)). *For a function f of bounded Hardy–Krause variation, the QMC error using an N -point low-discrepancy sequence is bounded by*

$$|\widehat{V}^{QMC} - V| \leq V(f) D_N^*, \quad D_N^* = O((\log N)^d / N),$$

essentially $O(1/N)$ up to logarithmic factors, versus $O(1/\sqrt{N})$ for random Monte Carlo.

Remark. QMC is asymptotically faster than plain Monte Carlo but requires “smooth” integrands. For payoffs with kinks (a vanilla call at the strike) the gain is modest; for smooth payoffs (average-strike Asians) the gain can be huge.

7.5 American options: Longstaff–Schwartz

Monte Carlo natively handles European options. American options require a *policy* choice at every node — an early-exercise rule — that cannot be found by forward simulation alone. The Longstaff–Schwartz algorithm (2001) fits this policy by regression:

Definition 7.3 (Longstaff–Schwartz algorithm). For an American option with payoff f_t :

1. Simulate N paths $S_0^{(i)}, \dots, S_T^{(i)}$ under $\mathbb{Q}$.
2. At maturity T , set the continuation value $V_T^{(i)} = f_T(S_T^{(i)})$.
3. Iterating backwards from $t = T - \Delta t$ to 0:
 - Regress the discounted future value $e^{-r\Delta t} V_{t+\Delta t}^{(i)}$ on a basis of functions of $S_t^{(i)}$ (typically polynomial), giving an estimated continuation value $\widehat{V}_t^{cont}(S)$.
 - Set $V_t^{(i)} = \max(f_t(S_t^{(i)}), \widehat{V}_t^{cont}(S_t^{(i)}))$.
4. Return $V_0 = \frac{1}{N} \sum_i V_0^{(i)}$.

Example 7.3 (Longstaff–Schwartz on an American put)

For an American put on a non-dividend stock, a quadratic regression basis $\{1, S, S^2\}$ typically suffices to price to $\sim 1\%$. The optimal exercise boundary is recovered as the level S^* at which immediate exercise equals continuation.

7.6 Sensitivities and Greeks by Monte Carlo

Estimating derivatives of V with respect to inputs (Greeks) requires care.

Definition 7.4 (Pathwise derivative). For a smooth payoff f , differentiate inside the expectation:

$$\partial_\theta V = \mathbb{E}^\mathbb{Q}[\partial_\theta f(S_T(\theta))].$$

Estimated by MC as $\frac{1}{N} \sum_i \partial_\theta f(S_T^{(i)}(\theta))$. Fast but requires f smooth (fails at the strike of a vanilla call).

Definition 7.5 (Likelihood ratio (score) method). For any payoff f , use the identity

$$\partial_\theta V = \mathbb{E}^\mathbb{Q}[f(S_T) \cdot \partial_\theta \log p(S_T; \theta)].$$

The score function $\partial_\theta \log p$ absorbs the derivative into the density; the payoff is not differentiated. Works for any payoff but has higher variance.

7.7 Exercises

Exercise 7.1

Estimate the value of a European call with $S_0 = 100$, $K = 100$, $r = 5\%$, $\sigma = 20\%$, $T = 1$ using Monte Carlo with $N = 10^4$. Compare with the Black–Scholes formula and the antithetic-variate estimator.

Solution

BS: $d_1 = (0.05 + 0.02)/0.2 = 0.35$, $\Phi(d_1) = 0.637$; $d_2 = 0.15$, $\Phi(d_2) = 0.560$; $C_{BS} = 100 \cdot 0.637 - 100e^{-0.05} \cdot 0.560 = 10.4506$. Plain MC with 10^4 paths typically gives $\hat{C} = 10.4 \pm 0.3$ (95% CI). Antithetic MC gives $\hat{C} = 10.44 \pm 0.2$: about 30% variance reduction because the call payoff is monotone in Z .

Exercise 7.2

For the Asian call with payoff $(\bar{S} - K)^+$ where $\bar{S} = \frac{1}{n} \sum_{i=1}^n S_{t_i}$: derive the corresponding geometric-average call, express its price in closed form, and use it as a control variate.

Solution

Geometric average: $S^g = (\prod S_{t_i})^{1/n}$. Under GBM, $\ln S^g$ is Gaussian with easily-computed mean and variance, so the geometric Asian call has a closed-form BSM-like formula (with adjusted volatility $\sigma\sqrt{(2n+1)/(6(n+1))}$ and adjusted drift). Using the geometric price as Y' in the control-variate estimator for the arithmetic price gives variance reduction factor ~ 50 –100.

7.8 Callouts

Note

Monte Carlo pricing is the workhorse of quantitative finance because it *scales* to problems that no closed-form or PDE method can touch: high-dimensional baskets, complex payoffs, multi-factor rate models.

Tip

Always report a Monte Carlo estimate with its standard error. A price of “10.45” from $N = 10^4$ paths is meaningless without knowing the CI is ± 0.3 (say).

! Important

For high-dimensional integrands, plain Monte Carlo is unbeatable in scaling with dimension, but *variance reduction* can beat it by a large constant. In production, always implement at least one variance-reduction technique.

⚠ Warning

QMC's superior convergence rate requires smoothness of the integrand — often destroyed by kinked payoffs and by high-dimensional effective dimensions (path-dependent options). Do not assume QMC always beats MC.

🔥 Caution

Naive finite-difference Greeks $((V(+h) - V(-h)) / (2h))$ with two independent Monte-Carlo runs) have *worse* variance than the value itself. Always use pathwise or common-random-number techniques.

7.9 Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 6 ([slides_6.pdf](#)). Standard reference: Paul Glasserman, *Monte Carlo Methods in Financial Engineering*, Springer (2003). Longstaff–Schwartz reference: Longstaff & Schwartz, *Valuing American options by simulation: a simple least-squares approach*, Rev. Fin. Studies 14, 113 (2001).

8 Volatility modelling

Black–Scholes assumes constant volatility. The volatility surface says otherwise. This chapter discusses the two main families of models that repair BSM: *local volatility* (Dupire) and *stochastic volatility* (Heston, SABR), together with their strengths, weaknesses, and calibration practice.

8.1 The implied-volatility surface

Definition 8.1 (Implied-volatility surface). The **implied-volatility surface** is the two-dimensional function $(K, T) \mapsto \sigma^{IV}(K, T)$ obtained by inverting the BSM formula on quoted option prices. In equity markets it is quoted vs *log-moneyness* $\ln(K/F)$ and time to maturity T ; in FX vs *delta*.

Empirical facts:

- **Skew** for equities: IV falls with strike (OTM puts more expensive than OTM calls, reflecting crash fears).
- **Smile** for FX: IV is symmetric around ATM with wings up.
- **Term structure.** Smile flattens with maturity (converging to a level slightly above ATM).
- **Non-parallel movements.** The whole surface moves; it does not translate rigidly.

Remark. Reproducing the surface at $t = 0$ is *necessary but not sufficient*. A good model must also reproduce the *dynamics* of the surface — how skew changes when spot moves, how ATM vol responds to level. This is what separates local vol (bad dynamics) from stochastic vol (good).

8.2 Local volatility (Dupire)

The idea: rather than assume σ constant, let it depend on both spot and time, $\sigma = \sigma_L(S, t)$. Under a chosen function σ_L , GBM becomes

$$dS_t = rS_t dt + \sigma_L(S_t, t) S_t dW_t.$$

Theorem 8.1 (Dupire’s formula). *Given a full arbitrage-free surface of European call prices $C(K, T)$, the unique local volatility function that matches it is*

$$\sigma_L^2(K, T) = \frac{2 \partial_T C + 2rK \partial_K C}{K^2 \partial_K^2 C}.$$

Proof sketch. Use the forward Kolmogorov equation for the risk-neutral density $\rho(K, T)$: $\partial_T \rho = -\partial_K(rK\rho) + \frac{1}{2} \partial_K^2(\sigma_L^2 K^2 \rho)$. Multiply by the payoff $(S_T - K)^+$ and integrate; identify boundary terms with $\partial_T C$ and internal terms with $\partial_K^2 C$.

Remark (Interpretation). The Dupire formula turns the pricing surface into a calibration surface: it says every arbitrage-free market has *one* local vol that reproduces it. In practice the formula is applied to interpolated/smoothed IV surfaces; the raw finite differences are extremely noisy.

8.3 Local volatility: weaknesses

Example 8.1 (Sticky-strike vs sticky-delta)

Local vol implies “sticky-strike” behaviour: as spot moves the IV at fixed strike stays constant. Real markets are closer to “sticky-delta”: as spot moves the whole smile moves with it. This mismatch means the local-vol *dynamics* are wrong even if its calibration is perfect at $t = 0$.

Forward-starting options (cliquets, resettable Asians) are the diagnostic where local-vol failures are dramatic: local vol under-prices these products by 30–50%. This is why stochastic volatility models were introduced.

8.4 Stochastic volatility: Heston

Definition 8.2 (Heston model). The **Heston model** is

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{v_t} S_t dW_t^{(S)}, \\ dv_t &= \kappa(\bar{v} - v_t) dt + \sigma_v \sqrt{v_t} dW_t^{(v)}, \end{aligned}$$

with $\text{corr}(dW^{(S)}, dW^{(v)}) = \rho$. The variance follows a mean-reverting CIR process.

Theorem 8.2 (Heston characteristic function). *The characteristic function of $\ln S_T$ under Heston is*

$$\phi(u) = \exp(C(u, T) + D(u, T)v_0 + iu \ln S_0),$$

with C, D solvable in closed form (Riccati ODEs). European call prices follow by Fourier inversion (Carr–Madan).

Remark. The Heston model is the standard *analytically-tractable* stochastic-volatility model. Its five parameters $(\kappa, \bar{v}, \sigma_v, \rho, v_0)$ are calibrated jointly to reproduce the surface. Correlation $\rho < 0$ generates the equity skew; σ_v controls the smile curvature.

8.5 Stochastic volatility: SABR

For FX and interest-rate applications, the *SABR* model is standard:

Definition 8.3 (SABR model).

$$\begin{aligned} dF_t &= \alpha_t F_t^\beta dW_t^{(F)}, \\ d\alpha_t &= \nu \alpha_t dW_t^{(\alpha)}, \end{aligned}$$

with correlation ρ . Parameter $\beta \in [0, 1]$ tunes between normal ($\beta = 0$) and log-normal ($\beta = 1$).

Theorem 8.3 (Hagan's SABR formula). *For SABR the implied volatility $\sigma^{IV}(K, T)$ is given approximately by*

$$\sigma^{IV}(K, T) \approx \frac{\alpha}{(FK)^{(1-\beta)/2} z(K, F)} \cdot [1 + O(T)],$$

with an explicit correction z in terms of $\ln(F/K)$. This lets calibration be done analytically without PDE or Monte Carlo — the reason SABR is universal in swaptions markets.

8.6 Rough volatility (2010s)

A more recent line of work (Gatheral, Jaisson, Rosenbaum, 2014) argues volatility is *rougher* than a diffusion — the log-vol process has Hurst exponent $H \approx 0.1$, i.e. is closer to fractional Brownian motion than to standard BM.

Remark (Rough Bergomi model). The **rough Bergomi** model

$$d \ln v_t = \eta \int_0^t (t-s)^{H-1/2} dW_s^{(v)}$$

fits the SPX volatility surface with only three parameters. Rough vol is the state of the art but hard to simulate; Monte-Carlo methods with clever variance reduction are the current tool.

8.7 Calibration

Real-world model calibration:

Definition 8.4 (Calibration). **Calibration** is the process of choosing model parameters θ to minimise the pricing error on a chosen set of instruments:

$$\theta^* = \arg \min_{\theta} \sum_i w_i (C^{model}(K_i, T_i; \theta) - C^{mkt}(K_i, T_i))^2.$$

Weights w_i typically scale inversely with quote widths and vega.

Standard pitfalls: over-fitting to too many strikes; parameter instability from day to day; local minima in the objective. Practical calibration workflows use good initial guesses (e.g. ATM-quoted vols), staged optimisation (fit ATM first, then smile), and regularisation.

8.8 Exercises

Exercise 8.1

For the Heston model with $\kappa = 2$, $\bar{v} = 0.04$, $\sigma_v = 0.3$, $\rho = -0.7$, $v_0 = 0.04$, and $S_0 = 100$, $r = 3\%$, $T = 1$, simulate 10^4 paths via Euler scheme (with reflection to keep $v \geq 0$) and price a European ATM call. Compare with the Fourier-inversion price.

Solution

Euler with $\Delta t = 1/252$ and reflection ($v \rightarrow |v|$) gives $\hat{C} \approx 6.78 \pm 0.15$. The exact price (Fourier) is $C^{Heston} \approx 6.82$. The reflection scheme has a small bias due to the CIR boundary at zero; a Milstein

or QE scheme (Andersen 2008) is preferred for accuracy.

Exercise 8.2

Compute the local-volatility function from a simple parametric IV surface $\sigma^{IV}(K, T) = 0.2 - 0.1(\ln(K/F))$ (linear skew). What is $\sigma_L(K = F, T)$?

Solution

Directly applying Dupire: at $K = F$ the surface has $\sigma^{IV} = 0.2$; the skew derivatives combined with the Dupire formula give $\sigma_L^2 \approx (0.2)^2 + O(\text{skew}^2)$, so $\sigma_L(F, T) \approx 0.2$. Away from the money, σ_L is roughly twice the slope of the IV skew: this “ $\sigma_L \approx 2\sigma^{IV}$ ” rule is Derman’s classical mnemonic for rough estimation.

8.9 Callouts

Note

Local vol and stochastic vol are complementary: local vol calibrates the smile perfectly but has wrong dynamics; stoch vol has right dynamics but doesn’t calibrate perfectly. Practitioners often use a *hybrid* local–stochastic vol (LSV) model to get both.

Tip

When calibrating Heston, start from the ATM term structure to fit $\bar{v}, \kappa, v_0$; then fit the smile to determine σ_v, ρ . Trying to fit everything jointly is unstable.

Warning

Stochastic vol models can produce numerically-negative variance in Euler simulations. Always use variance-preserving schemes (Andersen QE, exact simulation of CIR) for production.

Important

Implied vol is a *quote*, not a physical parameter. Modelling it directly (SVI, SSVI, rough Bergomi in IV space) is often more efficient than modelling the underlying process.

8.10 Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 7 ([slides_7.pdf](#)). Jim Gatheral, *The Volatility Surface: A Practitioner’s Guide*, Wiley (2006). Original references: Heston, *A closed-form solution for options with stochastic volatility*, Rev. Fin. Studies **6**, 327 (1993); Hagan et al., *Managing smile risk*, Wilmott (2002).

9 Interest rate modelling

Modelling interest rates is at once the oldest and the most subtle problem in derivative pricing. Rates are the numéraire, so their model *matters even if you only price equity options*. This chapter surveys the classical one-factor short-rate models, the HJM framework, LIBOR/SOFR market models, and the post-2008 multi-curve reality.

9.1 The short rate

Definition 9.1 (Short rate). The **short rate** r_t is the annualised interest rate applicable to an instantaneous loan at time t . The money-market account grows as $dB_t = r_t B_t dt$, so $B_t = \exp(\int_0^t r_s ds)$.

Definition 9.2 (Zero-coupon bond). The **zero-coupon bond** $P(t, T)$ is the value at time t of receiving 1 at time T . Under the risk-neutral measure with numéraire B ,

$$P(t, T) = \mathbb{E}^{\mathbb{Q}}[e^{-\int_t^T r_s ds} \mid \mathcal{F}_t].$$

The **yield** is $y(t, T) = -\frac{1}{T-t} \ln P(t, T)$; the **forward rate** is $f(t, T) = -\partial_T \ln P(t, T)$.

9.2 Vasicek and CIR models

The two classic one-factor short-rate models:

Definition 9.3 (Vasicek (1977)). The **Vasicek model** posits an Ornstein–Uhlenbeck short rate under $\mathbb{Q}$:

$$dr_t = \kappa(\theta - r_t) dt + \sigma dW_t^{\mathbb{Q}}.$$

The short rate is Gaussian and mean-reverting; unfortunately it can go negative.

Proposition 9.1 (Vasicek zero-coupon bond). *Under Vasicek,*

$$P(t, T) = A(t, T) \exp(-B(t, T)r_t),$$

with $B(t, T) = (1 - e^{-\kappa(T-t)})/\kappa$ and $A(t, T)$ an explicit function of κ, θ, σ . The forward-rate curve is

$$f(t, T) = e^{-\kappa(T-t)}r_t + \theta(1 - e^{-\kappa(T-t)}) - \frac{\sigma^2}{2\kappa^2}(1 - e^{-\kappa(T-t)})^2.$$

Definition 9.4 (Cox–Ingersoll–Ross (1985)). The **CIR model** solves the negative-rate problem with a square-root diffusion:

$$dr_t = \kappa(\theta - r_t) dt + \sigma\sqrt{r_t} dW_t^{\mathbb{Q}}.$$

For $2\kappa\theta \geq \sigma^2$ (Feller condition) the process stays strictly positive. Zero-coupon bond prices are still affine in r_t .

Remark. Since 2015 several major currencies (EUR, JPY, CHF) have exhibited *negative* short rates. The Vasicek model, once considered defective for allowing negative rates, is now correct in this respect; the CIR model, once considered “safe”, is inconsistent with observed data.

9.3 The HJM framework

Rather than model the short rate, Heath–Jarrow–Morton (1992) model the entire *forward-rate curve* $\{f(t, T) : T \geq t\}$ directly:

Definition 9.5 (Heath–Jarrow–Morton). Under HJM, the forward-rate curve evolves as

$$df(t, T) = \alpha(t, T) dt + \sigma(t, T)^\top d\mathbf{W}_t,$$

with $d\mathbf{W}_t$ a d -dimensional Brownian motion. HJM is the framework in which every interest-rate model can be stated.

Theorem 9.2 (HJM drift condition). *For the discounted bond prices to be martingales, the drift α is not free but determined by σ :*

$$\alpha(t, T) = \sigma(t, T)^\top \int_t^T \sigma(t, u) du.$$

This is the celebrated HJM drift condition: choosing the volatility structure σ fully specifies the model.

9.4 LIBOR market model

For pricing swaptions and caps, the natural quantity is not the instantaneous forward rate but the *forward LIBOR* $L(t; T_i, T_{i+1})$ over a discrete tenor.

Definition 9.6 (LIBOR Market Model (LMM)). Under the T_{i+1} -forward measure, each forward LIBOR $L_i(t)$ evolves as a log-normal:

$$dL_i(t) = \sigma_i(t) L_i(t) dW_t^{T_{i+1}}.$$

Different L_i are driven by correlated Brownians. The joint dynamics is determined by the volatility profile $\{\sigma_i\}$ and the correlation matrix.

Corollary 9.3 (Black’s formula for caplets). *Under LMM, a caplet payoff $(L(T_i; T_i, T_{i+1}) - K)^+$ is priced by Black’s formula:*

$$\text{Caplet}_0 = P(0, T_{i+1}) [L_i(0) \Phi(d_1) - K \Phi(d_2)] \cdot \delta_i,$$

with $d_{1,2}$ as in BSM but on the forward rate.

9.5 Post-2008: multi-curve

Before 2008, the market used a single “zero curve” for both discounting and forward-rate projection. The LIBOR-OIS basis blew out during the crisis (LIBOR reflects bank credit risk; OIS is closer to risk-free), and the market switched to a *multi-curve* framework:

Definition 9.7 (Multi-curve framework). • **Discount curve.** Built from OIS swap rates; used to discount all cash flows.

• **Projection curves.** One per LIBOR tenor (1M, 3M, 6M, 12M); used to project future coupons.

The two are related but not identical, and their basis (LIBOR – OIS) is stochastic and priced separately.

Remark (After LIBOR). LIBOR is being phased out globally (2021–2023). Replacement rates are: SOFR (USD), €STR (EUR), SONIA (GBP), SARON (CHF), TONAR (JPY). Each is an *overnight* rate averaged over accrual periods — a very different animal from term LIBOR. The transition has forced a repricing of trillions of dollars of derivatives.

9.6 RFRs and backwards-looking rates

Definition 9.8 (Risk-free rate (RFR)). An **RFR** is an overnight interest rate benchmark based on actual transactions (SOFR is the volume-weighted median of overnight repos on US Treasury collateral). Unlike LIBOR, RFRs are *backward-looking*: the rate for a period $[T_{i-1}, T_i]$ is

$$R_i = \frac{1}{T_i - T_{i-1}} \int_{T_{i-1}}^{T_i} r_s ds.$$

The rate is not known until T_i .

Example 9.1 (SOFR compounding)

A SOFR-based coupon over 3 months compounds daily overnight rates:

$$R^{comp} = \prod_{k=1}^n (1 + r_{t_k} \delta_k) - 1,$$

paid at the end of the accrual period. The uncertainty until the last day makes hedging non-trivial and forces a partial *lookback* convention (paying a couple of days after T_i) in practice.

9.7 Exercises

Exercise 9.1

Derive the ZCB price formula for the Vasicek model by direct calculation of $\mathbb{E}^Q[\exp(-\int_t^T r_s ds)]$.

Solution

Since r_s is Gaussian under Vasicek, $\int_t^T r_s ds$ is Gaussian with mean $\mu(T-t) + (r_t - \mu)B$ and variance $\sigma^2 \int_t^T B(s, T)^2 ds$. The expectation of exp of a Gaussian equals exp of mean plus half variance. Substitute and identify with the affine form.

Exercise 9.2

Show that in a flat, upward-sloping yield curve $y(T) = y_0 + sT$ the forward rate is $f(T) = y_0 + 2sT$ — twice the slope of the yield curve.

Solution

$P(0, T) = e^{-y(T)T} = e^{-(y_0 T + sT^2)}$, so $-\ln P = y_0 T + sT^2$; forward rate $f(T) = \partial_T(y_0 T + sT^2) = y_0 + 2sT$. This is the “forward slope is twice the yield slope” rule.

9.8 Callouts

i Note

Short-rate models are elegant but ill-suited to modern rate markets: they cannot fit the volatility surface of caps and swaptions jointly. LMM or HJM is required in practice.

💡 Tip

When calibrating a rate model, always match the yield curve *exactly* first; a rate model that misprices vanilla bonds is worthless. Sensitivity to skew is a distant second-order concern.

! Important

The LIBOR transition is *the* biggest single change to derivative markets since Black–Scholes. Any code that still hard-codes LIBOR discounting is broken; every derivative pricer had to be rewritten in 2021–2023.

⚠ Warning

Negative rates break the CIR and log-normal LMM models. If you’re going to price European rate options in a negative-rate regime, use a normal (Bachelier) model on the forward rate rather than a log-normal.

9.9 Sources

Prof. Fontana, *Stochastic Finance* slides, Chapter 8 ([slides_8.pdf](#)) and the practical [RFR_Practical.pdf](#). Musiela & Rutkowski, *Martingale Methods in Financial Modelling*, Chapters 11–12, 15. Post-LIBOR reference: Andersen & Piterbarg, *Interest Rate Modeling*, Vols. 1–3, Atlantic Financial Press (2010, updated 2021).